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Paleomagnetism of Some Rocks from the Mesaverde Group, Southwestern Wyoming and Northeastern Utah

ABSTRACT

The results of this investigation suggest that a viscous partial thermoremanent magnetization may be important for preserving paleomagnetic directions in some sedimentary rocks.

INTRODUCTION

A paleomagnetic study of Upper Cretaceous (Campanian) sedimentary rocks of southwestern Wyoming and northeastern Utah demonstrates that rocks of weak intensity can be measured and useful information extracted from the data. Four different stratigraphic sections were sampled over a vertical interval whose location is shown on Figure 1, and the stratigraphy is summarized in Figure 2.

Three magnetically oriented drill cores were collected stratigraphically at each 60-m vertical interval or as close to that interval as possible. Each core was cut into three specimens of length equal to its diameter in the laboratory. An astatic magnetometer was used to measure the directions and intensity of each specimen.

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RESULTS

The site mean directions of the NRM are listed in Table 1. The NRM declinations and inclinations measured with respect to the core axis were first restored to their *in situ* position and then rotated to the horizontal about the strike of the beds, a horizontal fold axis being assumed. The mean intensity of magnetization for 427 specimens is 1×10^{-5} emu/cc while the susceptibility of selected specimens varies from 1 to 2×10^{-6} emu/cc.

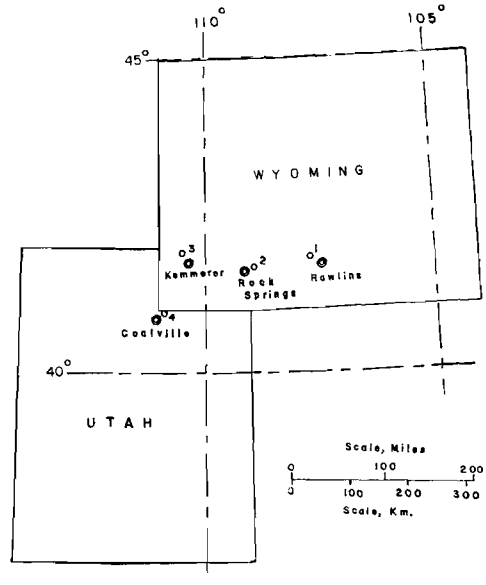


Figure 1. Index map of Wyoming and Utah showing location of collecting sites: (1) Point of Rocks section, (2) Superior section; (3) Ham's Fork section; (4) Echo Canyon section.

Storage tests suggest that no viscous component is present as the remeasurement of a suite of 20 specimens from a single site indicates (Table 2). The remeasured specimens were stored in a magnetically field-free area for one year with the north direction of the orientation mark to the south and inverted. The difference between the mean directions before and after storage is statistically insignificant, indicating the scatter of the data is not caused by soft components of magnetization.

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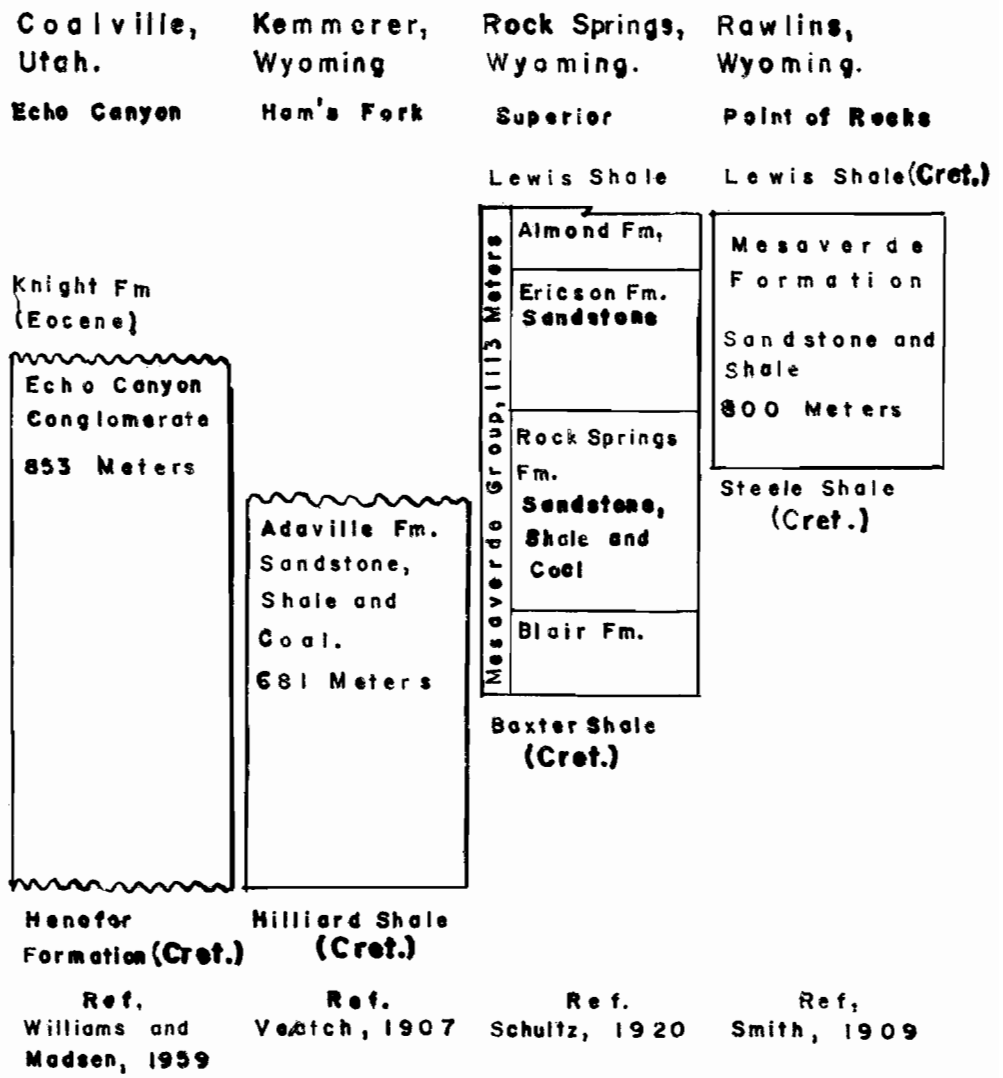


Figure 2. Cross section showing stratigraphic relationships of rock units sampled.

Thermal demagnetization studies (Irving and others, 1961) were made to determine stability of magnetization of the rocks sampled. The data for a suite of specimens demagnetized by stages is listed in Table 3. Although directions of individual specimens moved to some degree, the mean direction did not change significantly. While these experiments indicate that randomly distributed components are thermally unstable, the site mean of the NRM appears to give a good estimate of a stable mean direction.

Additional thermal demagnetization experiments were performed to determine if the type of magnetization present could be ascertained. Figure 3 indicates that the magnetization is thermally distributed (Irving and Opdyke, 1965) and that all measureable magnetization is lost on heating the core sample to a temperature of 300° C; therefore, a broad blocking temperature spectrum is implied.

Demagnetization curves for an artificially created TRM are shown in Figure 4. Only a small amount of intensity is lost to a temperature of 300° C, the bulk having been acquired above that temperature.

The demagnetization of a partial TRM is shown in Figure 5. These specimens were

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TABLE 1. SITE MEAN DIRECTIONS FOR EACH COLLECTING AREA

Point of Rocks Section						Superior Section					
Site	N	D_m	I_m	k	R	Site	N	D_m	I_m	k	R
3	12	161.8	76.5	1.93	6.3086	31	9	347.8	86.4	6.83	7.8282
9	9	129.6	78.4	2.95	6.2842	33	9	10.5	56.6	2.28	5.4932
11	9	249.1	59.7	2.25	5.4443	46	6	0.2	53.9	5.05	5.0101
22	9	305.5	59.6	2.51	5.8078	55	9	21.0	4.4	1.53	3.7754
32	9	252.7	69.7	2.10	5.1875	57	9	353.8	36.2	1.70	4.3008
34	7	258.6	76.6	1.45	2.8728	63	9	141.7	61.2	1.57	3.9147
40	9	252.8	33.4	2.27	5.4752	67	8	351.1	69.0	1.53	3.4153
46	9	287.2	56.1	6.94	7.8478	72	9	334.6	27.0	1.63	4.0938
48	9	224.0	46.0	2.25	5.4389	75	9	356.1	56.7	3.47	6.6960
55	8	185.8	20.0	9.95	7.2967	78a	8	36.2	32.8	8.92	7.2150
59	9	281.8	31.2	10.8	8.2580	78b	12	15.6	18.3	2.95	8.2649
67	7	273.8	39.3	28.5	6.7893	80	9	335.9	71.5	2.26	5.4601
69	9	276.6	32.8	30.9	8.7415	82	9	291.6	24.5	1.54	3.8032
82	9	57.5	69.5	2.71	6.0444	85	8	45.3	69.4	4.30	6.3713
86	20	232.3	32.8	15.0	18.7300	92	9	24.3	38.2	5.24	7.4734
						96	9	57.7	75.3	2.69	6.0275
						97	21	272.0	62.6	1.82	10.0357
Ham's Fork Section						Echo Canyon Section					
1	20	8.7	45.1	2.04	10.6860	2	9	296.3	57.6	3.23	6.5257
8	4	178.9	47.0	6.38	3.5296	4	4	250.5	−23.7	1.39	1.8430
10	2	56.8	23.9	1.34	1.3387	6	20	19.5	40.7	4.90	16.1007
14	8	210.6	66.4	2.28	4.9244	8	1	147.2	29.5	..	..
17	8	0.6	62.9	3.85	6.1832						
18	8	269.7	75.8	1.49	3.3131						
22	9	140.2	65.4	2.27	5.4805						
24	9	38.2	37.8	2.85	6.1947						
27	5	347.2	43.6	5.01	4.2018						

N, number of specimens averaged
D_m, mean declination
I_m, mean inclination (positive unless otherwise indicated)
k, Fisher's precision parameter (Fisher, 1953)
R, vector resultant of N unit vectors

TABLE 2. MEAN NRM DIRECTION FOR A SINGLE STRATIGRAPHIC UNIT (UNIT 86, POINT OF ROCKS SECTION) BEFORE AND AFTER STORAGE IN A FIELD-FREE AREA FOR ONE YEAR

	N	D_m	I_m	$\alpha_{(95)}$	k
Before Storage	20	232.3°	32.8° (down)	8.73°	15.0
After Storage	20	231.2°	40.5° (down)	7.75°	18.7
Difference		1.1°	7.7°		

N, number of specimens
D_m, mean declination
I_m, mean inclination (positive)
$\alpha_{(95)}$, confidence cone of 95 percent around mean direction of magnetization
k, estimate of precision parameter

thermally demagnetized before remagnetization. The intensity of PTRM is almost identical to that of the original NRM, and the shapes of the curves are similar. These results suggest that the magnetization of these rocks is most probably a viscous partial TRM (Briden, 1965) imposed during burial by later sedimentation, and fixed at the time of the first uplift of the area at the close of the Cretaceous Period.

A pole position calculated from the data of Table 4 is located at 65.1° N., 162° W. This is a revision of data previously reported (Kilbourne, 1967). An intensity weighting system was used to calculate this pole position by

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TABLE 3. RESULTS OF THERMAL DEMAGNETIZATION OF A SUITE OF REPRESENTATIVE SPECIMENS

Temperature	N	D_m	I_m	k	$\phi_{(95)}$
20° C.	20	3.2	74.7	2.0	32.8
105° C.	20	3.9	45.2	3.3	21.6
152° C.	19	359.3	48.5	4.1	18.9
200° C.	20	357.1	46.2	9.1	11.4

N, number of specimens
D_m, declination east of north
I_m, inclination positive downward
k, estimate of precision parameter
$\alpha_{(95)}$, radius of cone of confidence

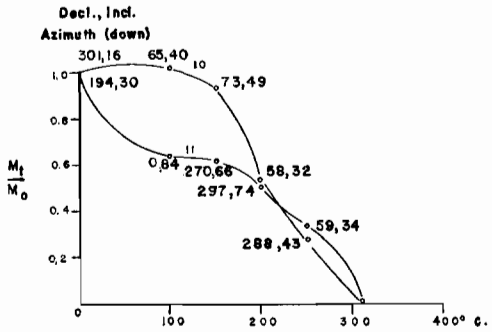


Figure 3. Thermal demagnetization curves of NRM for representative specimens (heating only). M_t, intensity of magnetization after heating to temperature t. M_o, original intensity of magnetization.

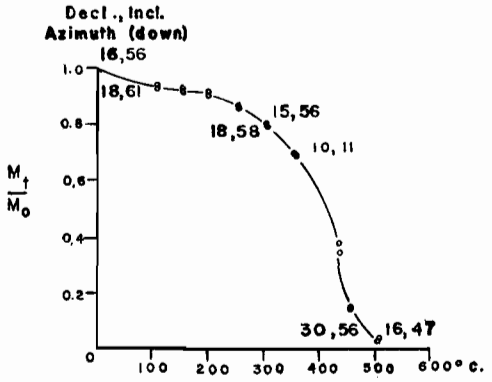


Figure 4. Thermal demagnetization curves (heating only) for representative specimens of a TRM imposed by cooling from a temperature of 680° C in the earth's field (0.56 Gauss). Symbols as in Figure 3.

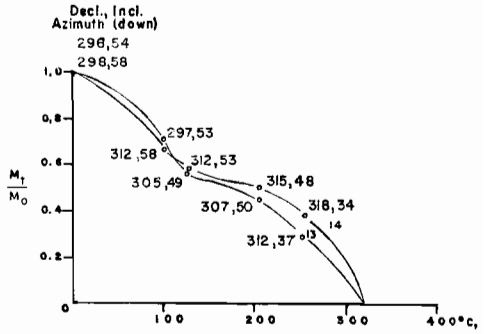


Figure 5. Thermal demagnetization curves (heating only) for representative specimens of a partial TRM imposed by cooling from a temperature of 322° C in the earth's field (0.56 Gauss). Symbols as in Figure 3.

which specimens of greater intensity are given more weight than weaker specimens. This is accomplished by multiplying the direction cosines of each specimen by its intensity. The pole positions previously reported in the literature for rocks of Cretaceous age are listed in Table 5 for comparative purposes.

CONCLUSIONS

This study suggests that paleomagnetic data can be preserved in sedimentary rocks of weak intensity as a viscous partial thermoremanence and that, while the data may be widely scattered, useful information can be extracted from it. The pole position calculated from data obtained from these Upper Cretaceous sedimentary rocks does not differ significantly from that determined from igneous rocks of similar

TABLE 4. AVERAGE DIRECTION OF MAGNETIZATION AND PALEOMAGNETIC POLE POSITION FOR MESAVERDE GROUP

	Direction of Magnetization					Paleomagnetic Pole	
N	D_m	I_m	k	R_m	$\alpha_{(95)}$	λ	ϕ
45	326.7	68.8	3.6	32.8	13.2°	65.1° N.	162.0° W.

Symbols identified in Table 3, except for R_m, vector resultant of N unit vectors

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TABLE 5. VIRTUAL GEOMAGNETIC POLES REPORTED FOR THE CRETACEOUS OF NORTH AMERICA
(Table modified *from* Larochelle, 1968)

Rock Unit	Pole Position	δm	δp
Mt. Megantic SE Quebec	68° N., 171° E.	6°	9°
Isachaen diabase Canadian Arctic	69° N., 180°	14.1°	13.6°
Sierra Nevada Batholith, California	69° N., 165° W.		
Mt. Ascutney Int. SE Vermont	65° N., 173° W.		
Elkhorn Mountains Volcanics, SW Montana	69° N., 171° W.	10°	7°
Monteregian Hills SE Quebec	69° N., 173° W.	5°	3.7°
Mesaverde Formation SW Wyoming	65° N., 162° W.	23.8°	21.5°

δp, semi-axis of elliptical error (along great circle between paleopole and site)
δm, semi-axis of elliptical error (at right angles to δp)

age. All of the circles of confidence or ovals of confidence reported in the literature for rocks of Cretaceous age from North America overlap to some degree.

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